

SCHEMA DESIGN FOR AMAZONE ONLINE SERVICE

// 1. Customers Collection

```
customers: {
  _id: ObjectId,
  customer_id: string,
  name: string,
  gender: string,
  age: int,
  address: [{ //embedded address details
    house_number: string,
    street: string,
    city: string,
    postcode: string
  }],
  orders: [ref<orders.order_id>], // reference to orders
  ratings: [{ // ratings by the customer
    product: ref<product._id>, // Reference to product
    rating: float // rating score
  }],
  recommendations: [ref<product._id>] // recommended products
};
```

// 2. Products Collection

```
products: {
  _id: ObjectId,
  name: string, // product name
  description: string, // brief description
  dimensions: string, // dimensions of product
  price: float, // price for customers
  cost: float, // cost to Amazone
  avg_rating_score: float
};
```

// 3. Orders Collection

```
orders: {
  _id: ObjectId,
  total_cost: float, // total cost to Amazone
  is_current: boolean, // true = current, false =
past
  order_date: Date,
  order_items: [{
    product: ref<product._id>, // Reference to product
    quantity: int // qty of product in order
  }],
  customer_id: ref<customer._id>, // customer who placed order
};
```

// 4. Fresh Products Collection

```
fresh_products: {
  _id: ObjectId,
  categories: string,          //category(e.g.,bakery,drinks,
etc)
  weight_or_quantity: float,    // weight or quantity
  expiry_date: date,
  origin_country: string,
  product: ref<product._id>,    // reference to product
  store_id: ref<stores._id>     // reference to the store
};
```

// 5. Other-Products Collection

```
other_products: {
  _id: ObjectId,
  shipping_weight: float,       // shipping weight of product
  product: ref<product._id>    // Reference to product
};
```

// 6. Other Products - Books Collection

```
books: {
  _id: ObjectId,
  author: string,
  publisher: string,
  publication_year: Date,
  ISBN: string,
  other_product: ref<other_product._id> //references other
product
};
```

// 7. Other Products - CDs Collection

```
cds: {
  _id: ObjectId,
  artist: string,
  number_of_tracks: int,
  total_playing_time: int,
  publisher: string,
  other_product: ref<other_product._id> //references other
product
};
```

// 8. Other Products - Mobile Phones Collection

```
mobile_phones: {
  _id: ObjectId,
  brand: string,
  model: string,
  colour: string,
```

```
    features: string,
    other_product: ref<other_product._id>    //references other
product
};
```

// 9. Other Products - Home Appliances Collection

```
home_appliances: {
    _id: ObjectId,
    colour: string,
    voltage: string,
    style: string,          // product style
    other_product: ref<other_product._id>    //References other
product
};
```

// 10. Amazone Partner Stores Collection

```
stores: {
    _id: ObjectId,
    store_id: string,
    address: string,          // store address
    location: {               // geolocation of store
        "type": "Point",
        "coordinates": [float, float] //Longitude & Latitude
    },
    grocery_items: [{        // list of grocery items in store
        item: ref<fresh_product._id>,
        inventory: int
    }]
};
```

// 11. Amazone Partner Drivers Collection

```
divers: {
    _id: ObjectId,
    driver_id: string,
    name: string,
    gender: string,
    age: int,
    address: string,
    current_location: {
        "type": "Point",
        "coordinates": [float, float] //Longitude & Latitude
    },
    status: string,
    jobs: [{
        order: ref<orders._id>,
        job_status: string, // 'picked', 'in-transit', 'delivered',
        distance_traveled: float, //km
    }]
};
```

```
        pay_out: float
    }],
    total_deliveries: int,
    total_pay_out: float,
};
```

// 12. Daily Inventories Collection

```
daily_inventories: {
    _id: ObjectId,
    date: Date,                // date of inventory level
    quantity: int,            // available quantity
    storage_warehouse: {      // storage information
        name: string,         // warehouse name
        location: {           // geolocation
            "type": "Point",
            "coordinates": [float, float] //Longitude & Latitude
        }
    },
    product: ref<product._id> // Reference to Product
};
```